

Recall from last class:

- For a matrix A define A_{ij} to be the matrix obtained from A by deleting row i and column j .
- For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{n+1} a_{1n} \det(A_{1n}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}).$$

We call this the cofactor expansion across the first row.

- **Theorem 3.1:** Cofactor expansion across any row or column is the same, and hence each cofactor expansion gives the determinant.
- Let $A = [a_{ij}]$ be an $n \times n$ matrix, then A is **upper triangular** if $a_{ij} = 0$ when $i > j$.
- **Theorem 3.2:** Let A be an $n \times n$ upper triangular matrix, then

$$\det(A) = a_{11} a_{22} \cdots a_{nn} = \prod_{j=1}^n a_{jj}.$$

1. Compute the determinant of $C = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}$.

Definition 1. Let $A = [a_{ij}]$ be an $n \times n$ matrix, then A is **upper triangular** if $a_{ij} = 0$ when $i > j$.

Theorem 3.2: Let A be an $n \times n$ upper triangular matrix, then

$$\det(A) = a_{11}a_{22} \dots a_{nn} = \prod_{j=1}^n a_{jj}.$$

2. Prove the theorem.

Goal: Prove $\det(A) \neq 0$ if and only if A is invertible.

Theorem 3.3: If A is an $n \times n$ matrix and E is an elementary matrix, then

$$\det(EA) = \det(E) \det(A),$$

and

$$\det(E) = \begin{cases} 1 & \text{if } E \text{ represents adding a multiple of a row to another row,} \\ -1 & \text{if } E \text{ represents interchanging two rows,} \\ r & \text{if } E \text{ represents scaling a row by a factor of } r. \end{cases}$$

3. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad E_1 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 10 & 0 \\ 0 & 1 \end{bmatrix}.$$

Check the theorem for these matrices.

4. Prove the theorem.

Theorem 3.6: If A and B are n by n matrices, then

$$\det(AB) = \det(A) \det(B).$$

5. Prove the theorem.

6. **Corollary:** A is invertible if and only if $\det(A) \neq 0$. Prove the corollary.

7. True or False:

(a) If A is an $n \times n$ invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

(b) If A is an $n \times n$ matrix, then $\det(A) = \det(A^T)$.

(c) If $A \sim B$, then $\det(A) = \det(B)$.

8. We can think of $\det$ as a function from the set of $n \times n$ matrices to the real numbers. Is this function linear?